MILITARY SPECIFICATION

IGNITER, MARK 247 MOD 0

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope.- This specification covers the requirements for an Igniter, Mk 247 Mod 0, a cartridge used as an igniter for solid propellant gas generators.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-7105	Pipe Threads, Taper, Aeronautical, National Form, Symbol ANTP

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-304	Temperature and Humidity Test for Use in Development of Fuses

FSC 1420

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STANDARDS

Military (continued)

MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MS33586	Metals - Definition of Dissimilar

DRAWINGS

Bureau of Naval Weapons

1617695	Guided Missile Container, Mark 213 Mod 0
2043442	Igniter, Mk 247 Mod 0 Specification Control Drawing

PUBLICATIONS

Bureau of Naval Weapons

OP-2979	Terrier/Tartar Shipping and Handling System

Air Force-Navy Aeronautical Bulletin

No. 143	Specifications and Standards, Use of

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as specified by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

National Bureau of Standards

Handbook H28	Screw Thread Standards for Federal Services

Code of Federal Regulations

49CFR 71-78	Transportation, Interstate Commerce Commission Explosives and Other Dangerous Articles

(Application for copies shall be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publications should cite the latest edition and supplements thereto.)

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3. REQUIREMENTS

3.1 Preproduction samples.- Unless otherwise specified in the contract or order, preproduction sample igniters shall be manufactured by the methods and procedures proposed for the production item. These samples shall be tested as specified in Section 4 and are for the purpose of determining that production igniters meet the requirements of this specification.

3.2 General requirements.-

3.2.1 Design.- The igniter specified herein shall be designed to satisfy the requirements of this specification and Bureau of Naval Weapons Drawing 2043442.

3.2.2 Materials, parts, and processes.- Materials, parts, and processes used in the design and construction of this igniter shall be selected in accordance with the requirements of ANA Bulletin No. 143. However, the manufacturer may incorporate any new or improved materials, parts or processes, providing such materials, parts, and processes are equal to or superior in quality to those required by ANA Bulletin No. 143, if the incorporation of such materials, parts, or processes will result in a design improvement, an expedited delivery schedule, or a cost reduction without impairing the operation of the equipment specified herein. When non-standard materials, parts, or processes are used, the corresponding AN or MIL material, part, or process number(s) shall be referenced on the drawings and parts lists, if these equivalents may be used with no alteration and with no degradation of the requirements specified herein.

3.2.3 Metals.- Metals shall be of a corrosion-resistant type or shall be adequately protected to resist corrosion during the normal storage and service life of the igniter.

3.2.3.1 Dissimilar metals.- Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MS33586 shall not be used in intimate contact.

3.2.4 Ignition materials.- The ignition materials used as the output load of this igniter shall consist of:

Material	Weight, Percent
Boron	23.7 ± 2.0
Potassium Nitrate	70.7 ± 2.0
Binder	5.6 ± 0.5
	100.0

3.2.5 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with respect to installation and performance.

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3.2.6 Gages.- The contractor shall provide himself with all gages required to insure that the unit meets the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order or requisition, the Government will furnish drawings of pertinent Naval Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not relieve the contractor of his responsibility in the design and manufacture of gages that may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Naval Final Inspection Gages.

3.2.7 Threads.- Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28 and all supplements thereof. The class or fit for threads shall be as specified on the applicable drawings, and the gaging practice shall be as set forth in the aforementioned Handbook H28.

3.2.8 Request for deviation.- Requests for deviations from this specification, applicable drawings, specifications, publications, materials, and processes specified herein, shall be addressed to the Bureau of Naval Weapons, Washington 25, D. C., via the government representative.

3.2.9 Conflicting requirements.- Conflicting requirements arising between this specification or any specification, publication or drawing listed herein shall be referred in writing to the Bureau of Naval Weapons or its appointed agent for interpretation and resolution (or correction).

3.3 Performance.- Igniters shall satisfy the following performance requirements when tested as specified in 4.3.

3.3.1 Operating temperature limits.- The igniter shall operate in conformance with the requirements of this specification at all temperatures between -75°F and 170°F, and shall withstand sudden temperature changes within the range of -75°F to 170°F.

3.3.2 Attitudes.- The igniter shall operate in conformance with the requirements of this specification regardless of the attitude of its supporting structure.

3.3.3 Dielectric strength.- The dielectric of the unloaded, unbridged igniter shall be capable of withstanding 1500 volts RMS, alternating current minimum applied between pins and between each pin and the igniter body without voltage breakdown.

3.3.4 Insulation resistance.- The pin-to-igniter body resistance of the loaded igniter shall be 2.0 megohms minimum at 500 volts direct current.

3.3.5 Circuit resistance.- The resistance between terminal pins of the igniter shall be 1.35 ± 0.20 ohms.

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3.3.6 Temperature and humidity.- The igniter shall operate in conformance with the requirements of this specification and shall complete the test sequences listed in Table 1 after being subjected to the temperature cycling and humidity test conditions specified in MIL-STD-304.

3.3.7 Maximum no-fire current.- The igniter shall not fire when a direct current of 0.40 (+0.01, -0.00) amperes is passed through the igniter circuit.

3.3.8 Minimum firing current.- Each igniter shall fire, as evidenced by rupture of the closure seal within 50 milliseconds, when a direct current of 1.5 amperes maximum is passed through the igniter circuit.

3.3.9 Pressure output.- The maximum pressure output of the igniter when the igniter has been conditioned to a temperature of 70° ±5°F and fired into a closed bomb chamber of 30 cubic centimeters free volume with configuration as shown in Figure 1 shall be 2000 ± 400 psig.

3.3.10 Firing time.- When the igniter is fired under conditions described in 3.3.9, the time from application of a direct current of 5 amperes to the time maximum pressure occurs shall be 45 ± 15 milliseconds.

3.3.11 Low temperature.- Each igniter, after conditioning at -75°F and upon application of a direct current of 1.5 amperes maximum across the bridge wires, shall function within 50 milliseconds as evidenced by rupture of the closure seal.

3.3.12 High temperature.- Each igniter, after conditioning at 170°F and upon application of a direct current of 1.5 amperes maximum across the bridge wires, shall function within 50 milliseconds as evidenced by rupture of the closure seal.

3.3.13 Post firing leakage.- After firing, the igniter shall not leak, blow out, or otherwise fail when subjected to an internal pressure of 3200 psig for a period of two minutes while at a temperature of 700°F minimum.

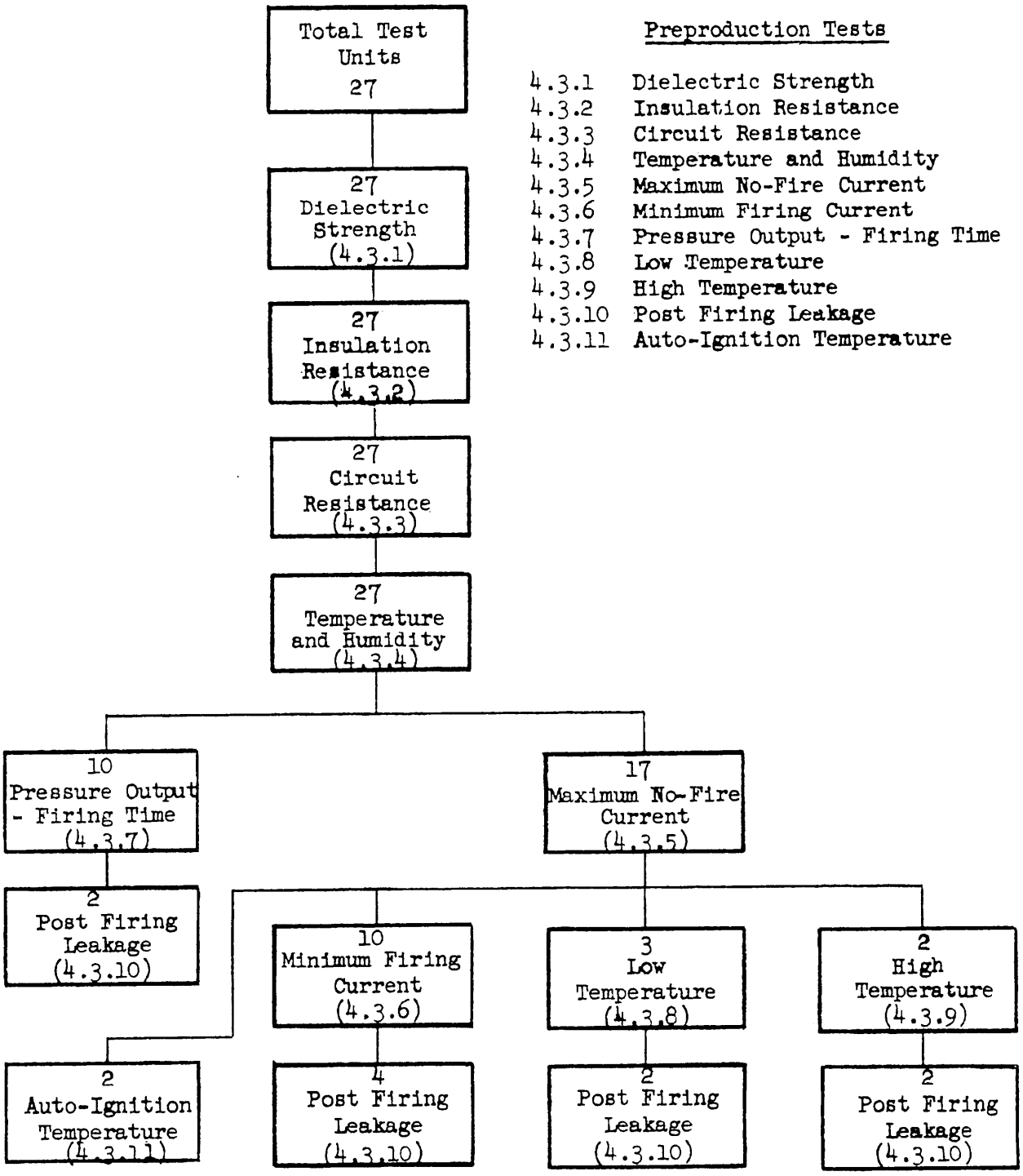
3.3.14 Auto-ignition temperature.- The igniter shall be capable of withstanding a temperature of 350°F for a period of eight hours without igniting, but shall not be required to function after being subjected to such environment.

3.4 Service life.- The igniter shall have a minimum service life equal to two years of actual installation and service, and shall be capable of storage for periods up to five years in dehydrated shipping containers at temperatures within the shipping containers from -65°F to 160°F without impairment of subsequent performance.

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TABLE I

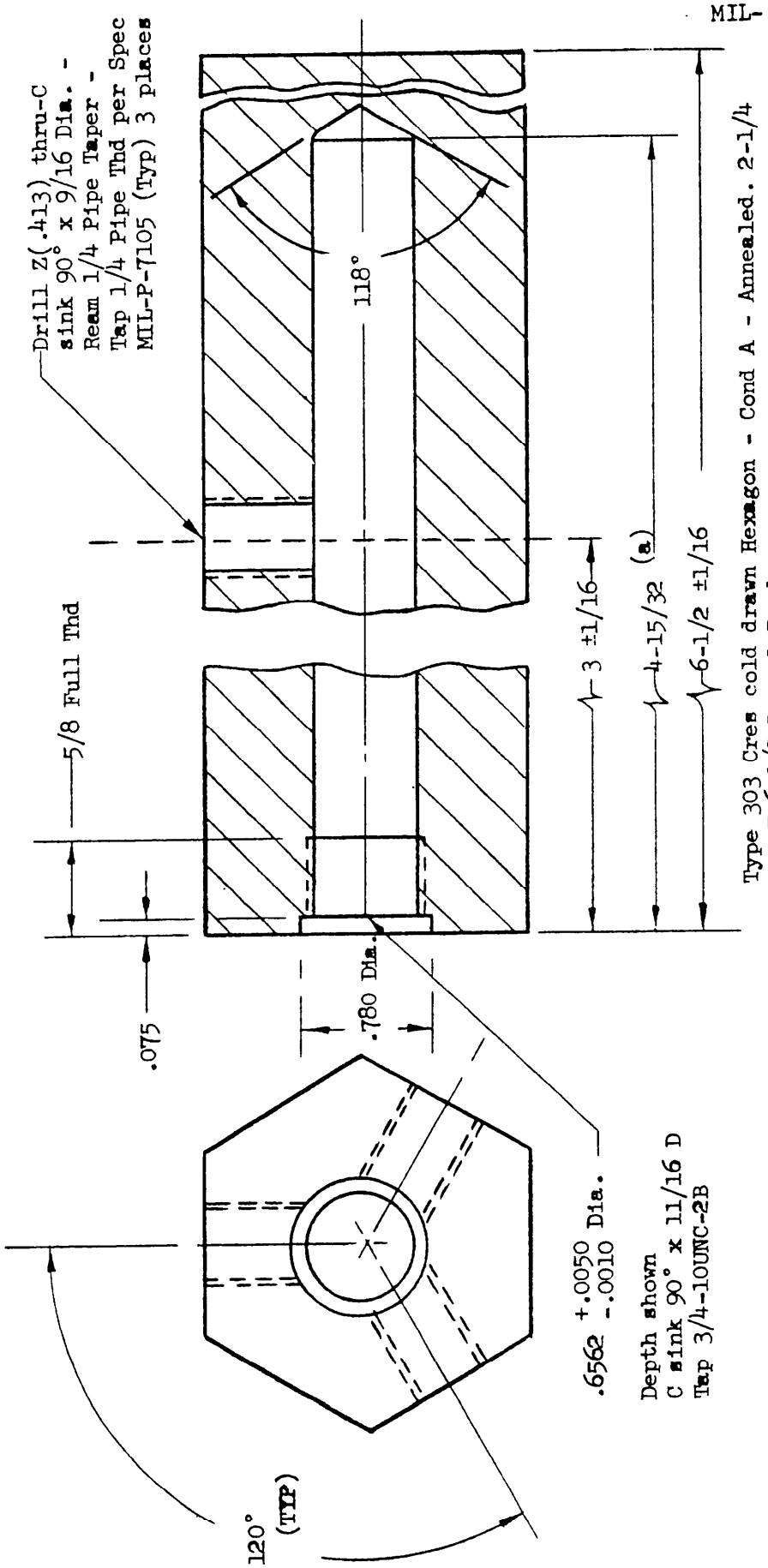
SCHEDULE FOR PREPRODUCTION TESTING



Preproduction Tests

4.3.1	Dielectric Strength
4.3.2	Insulation Resistance
4.3.3	Circuit Resistance
4.3.4	Temperature and Humidity
4.3.5	Maximum No-Fire Current
4.3.6	Minimum Firing Current
4.3.7	Pressure Output - Firing Time
4.3.8	Low Temperature
4.3.9	High Temperature
4.3.10	Post Firing Leakage
4.3.11	Auto-Ignition Temperature

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1. 125/ Machine Surface Roughness.
(a) May be increased to give free vol. of 30 cc with Igniter and Pipe Fittings in place.

Figure 1. Igniter Test Chamber, 30 cc.

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3.5 Identification.- Each igniter shall be permanently marked by etching or equivalent. Marking shall be in accordance with MIL-STD-130 and shall include the following:

Mark 247 Mod 0
Manufacturer's Lot Number
Manufacturer's Name or Trademark
Loaded Date: (Month and Year)

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government for the time period specified in the contract or order. The procuring activity reserves the right to perform or witness any of the inspections set forth in the specification or perform other tests to confirm compliance of the item with this specification.

4.2 Preproduction tests.- Prior to start of production, a lot of 27 igniters fabricated by the methods, techniques, and processes to be employed in production of igniters under this specification shall be subjected to and satisfactorily pass a preproduction test as specified in the following paragraphs. During the preproduction tests, no adjustments, repairs, or maintenance of the igniters shall be permitted unless required by this specification or otherwise approved by the procuring activity. The testing sequence shall be as shown in Table I and no deviation from this sequence shall be allowed.

4.3 Test methods.-

4.3.1 Dielectric strength.- Each unloaded igniter assembly shall be subjected to 1500 volts RMS, alternating current minimum applied between pins and between each pin and the igniter body for 10 ± 5 seconds. The test shall be conducted under atmospheric conditions of 77 ± 18°F, and a relative humidity of 70 ± 20 percent. Each igniter shall comply with the requirement of 3.3.3.

4.3.2 Insulation resistance.- The pin-to-igniter body resistance of each loaded igniter shall be measured when 500 volts direct current is applied between each pin and the igniter body. Each igniter shall comply with the requirement of 3.3.4.

4.3.3 Circuit resistance.- The resistance of the bridge wires shall be measured on each igniter using an Igniter Continuity Tester ALINCO Model 101-5a or equivalent. Each igniter shall comply with the requirement of 3.3.5.

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4.3.4 Temperature and humidity.- Each igniter shall be subjected to the temperature and humidity test specified in MIL-STD-304, except that the total test shall be limited to one JAN temperature and humidity cycle as defined in MIL-STD-304. Each igniter shall comply with the requirement of 3.3.6.

4.3.5 Maximum no-fire current.- A direct current of 0.40 (+0.01 -0.00) ampere shall be passed through the bridge wires of each igniter for 60 seconds minimum. Each igniter shall comply with the requirement of 3.3.7.

4.3.6 Minimum firing current.- Each igniter shall be subjected to a minimum firing current of 1.5 amperes maximum applied through the bridge wires. Each igniter shall comply with the requirement of 3.3.8.

4.3.7 Pressure output - firing time.- Each igniter shall be conditioned to a temperature of 70° ± 5°F and installed in a 30 cubic centimeter free volume closed bomb chamber having the configuration shown in Figure 1. A direct current of 5.00 ± 0.01 amperes shall be passed through the bridge wires. A continuous record shall be made showing the pressure versus time in the chamber from the time of application of the firing current to the time of attainment of maximum pressure. The pressure output and firing time of each igniter shall comply with the requirement of 3.3.9 and 3.3.10 respectively.

4.3.8 Low temperature.- Each igniter shall be conditioned to a temperature of -75° ± 3°F. While at this temperature a direct current of 1.5 amperes maximum shall be applied through the bridge wires of each igniter. Each igniter shall comply with the requirement of 3.3.11.

4.3.9 High temperature.- Each igniter shall be conditioned to a temperature of 170° ± 3°F. While at this temperature a direct current of 1.5 amperes maximum shall be applied through the bridge wires of each igniter. Each igniter shall comply with the requirement of 3.3.12.

4.3.10 Post firing leakage.- Each igniter shall be conditioned to a stabilized temperature of 700°F minimum. While at this temperature, the igniter shall be pressure tested with dry nitrogen for 2 minutes at 3200 (+100, -000) psig. A lead from the exposed end of the fired igniter will be carried through a water bath. Each igniter shall be examined for leakage as evidenced by gas displacement of water. Each igniter shall comply with the requirement of 3.3.13.

4.3.11 Auto-ignition temperature.- Each igniter shall be subjected to a temperature of 350° ± 3°F for a period of 8 hours. Each igniter shall comply with the requirement of 3.3.14. Igniters subjected to this test are unsuitable for use and, for reason of safety, shall be expended.

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4.4 Preproduction test report.- A preproduction report shall be prepared describing in detail the results of the preproduction tests. The report shall include a complete description of the test procedures and equipment used, including photographs where applicable. Copies of the original pressure-time records on closed chamber firings shall be included.

4.4.1 Approval of preproduction tests.- Approval of the preproduction tests shall be based on no performance failures and no mechanical defects in any igniter tested. However, should any igniter fail any one of the required tests, the Bureau of Naval Weapons may allow the contractor to test another igniter under the same conditions. If a second failure should occur, additional action on the part of the contractor, such as igniter modification or redesign, shall be subject to the approval of the Bureau of Naval Weapons.

4.5 Acceptance tests.- All igniters submitted for acceptance shall be subjected to the acceptance tests specified herein. Acceptance inspection of production lots shall be subject to surveillance by the Government inspector.

4.5.1 Lot.- A lot of igniters shall consist of igniters produced in a continuous production period with no change in process, components, or raw materials which are submitted for acceptance at the same time.

4.5.2 Sampling and inspection.-

4.5.2.1 Non-destructive examination and test.-

4.5.2.1.1 Each igniter in the inspection lot shall be examined to assure quality of workmanship, use of proper materials and assembly, and absence of visual defects or other imperfections that would effect the life or function of the igniter.

4.5.2.1.2 Dimensional requirements.- Each igniter in the inspection lot shall be examined for compliance with the dimensional requirements of applicable drawings.

4.5.2.1.3 Dielectric strength.- Sampling for compliance with the requirements of 3.3.3 shall be in accordance with MIL-STD-105, Inspection Level II. The sample shall meet an AQL of 1.5 when tested in accordance with 4.3.1.

4.5.2.2 Performance test sampling.- From each lot submitted for acceptance, the number of units to be tested shall be selected in accordance with MIL-STD-414, Inspection Level IV, normal inspection, with variability unknown-standard deviation method and subjected to the performance test specified herein.

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The plates shall be located in a position where they

4.5.2.2.1 Performance tests.-

4.5.2.2.1.1 Insulation resistance.- The pin-to-igniter body resistance of each igniter shall be measured in accordance with the procedure specified in 4.3.2. If any igniter fails to satisfy the requirements of 3.3.4, the lot shall be rejected.

4.5.2.2.1.2 Circuit resistance.- The resistance of the bridge wires of each igniter shall comply with the requirements of 3.3.5 to an AQL of 1.5 when measured in accordance with the procedure specified in 4.3.3. If the sample fails to meet the AQL requirement, the lot shall be subjected to 100 percent testing for resistance and units not meeting the requirements of 3.3.5 shall be rejected.

4.5.2.2.1.3 Maximum no-fire current.- Each igniter shall be subjected to the test specified in 4.3.5. If ignition of one or more igniters occurs, the lot shall be rejected.

4.5.2.2.1.4 Pressure output - firing time.- Each igniter shall be subjected to the test specified in 4.3.7. The maximum pressure and the time to maximum pressure of the sample shall meet the acceptability criterion as shown in MIL-STD-414 for an AQL of 1.5, with upper and lower specification limits as specified in 3.3.9 and 3.3.10 of this specification.

4.5.2.2.1.5 Post firing leakage.- Forty percent of the igniters tested in accordance with 4.5.2.2.1.4 shall be subjected to the test specified in 4.3.10. If any igniter leaks the lot shall be rejected.

4.5.3 Acceptance criteria.- If, during lot acceptance testing, any of the requirements of 4.5 are not met, the ignitor (see 4.5.2.1.1 and 4.5.2.1.2) or the lot, as applicable, shall be rejected.

4.5.3.1 Disposition of rejected lots.- The disposition of rejected lots shall be as specified in the contract or order.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- In addition to the following, Level A packaging shall be in accordance with the Code of Federal Regulations 49CFR71-78.

5.1.1.1 Cleaning, drying, and preservative application.- Not applicable.

5.1.1.2 Unit packaging.- Unit packaging of igniters shall be in accordance with BuWeps Drawing 1617695. The unit package shall contain 12 igniters.

5.1.1.3 Intermediate package.- Not applicable.

5.1.2 Level B.- Not applicable.

5.1.3 Level C.- Not applicable.

5.2 Packing.-

5.2.1 Level A.- In addition to the following, Level A packing shall be in accordance with the Code of Federal Regulations 49CFR71-78.

5.2.1.1 Exterior containers.- Igniters packaged as described in 5.1.1.2 shall be packed in exterior containers in accordance with OP 2979.

5.2.2 Level B.- Not applicable.

5.2.3 Level C.- Not applicable.

5.3 Marking.-

5.3.1 Special markings.- Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49CFR71-78.

5.3.2 Normal markings.- In addition to the special markings and the markings required by the contract or order, unit packages and exterior containers shall be marked in accordance with the requirements of MIL-STD-129.

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6. NOTES

6.1 General safety precautions.-

6.1.1 The loading, assembly, and handling of the explosive components of the igniters covered by this specification involve hazardous operations and therefore require the observance of suitable safety precautions. Use of this specification shall not be construed to relieve the contractor or the manufacturer of responsibility for the safety of his operations. Listed below are the minimum provisions which a contractor or manufacturer of the propellant should observe in order to fulfill his responsibility for safety. At Bureau of Naval Weapons, Navy Department, and other Government plants these provisions are mandatory.

6.1.1.1 All handling operations should be conducted in a neat and orderly manner.

6.1.1.2 Personnel handling the propellant should, insofar as practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion or deterioration from perspiration or other contaminating deposits.

6.1.1.3 The exposure of propellant materials should be controlled to minimize the absorption of moisture from the atmosphere or other sources during handling operations.

6.1.1.4 All explosives should be stored in suitable storage magazines located in accordance with the American Table of Distance (A. T. D.). All storage should be in adequate ready or service magazines, located in accordance with interplant distances.

6.1.1.5 Safe equipment and methods should be utilized for transporting and handling explosives.

6.1.1.6 Proper care should be exercised at all times to protect personnel from accidents, fires, or explosions and to limit damage to equipment. In this connection, the following precautionary measures should be observed.

6.1.1.6.1 Keep only minimum quantities of explosives present at each state of operation.

6.1.1.6.2 Keep propellant and explosive parts of the gas generator in approved, covered receptacles with covers in place when material is not being taken out or put into the receptacles. Where necessary, receptacles should be conductive in order to ground electrostatic charges.

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6.1.1.6.3 Protect all operations from electro-static charges by effectively grounding all machinery, equipment and fixtures and where necessary, provide suitable grounded conductive coverings on floors, work benches, tables, and workers' shoes.

6.1.1.6.4 Enforce, where necessary, the wearing of suitable footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning, and similar hazards.

6.1.1.6.5 Allow no fires or exposed sparking of electrical equipment and little or no inflammable material to be present in handling and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.1.1.6.6 Enforce good housekeeping and policing methods throughout the handling area and surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosive or inflammable material, and assure its removal from floors, walls, ceilings, ledges, tables, benches and work equipment. Clean up any spilled material immediately.

6.1.1.6.7 Employ properly proportioned and properly located protective barricades, screens, or shields at all required points.

6.2 Intended use.- The igniter covered by this specification is intended for use as an igniter for the Mark 2 and Mark 3 solid propellant gas generators.

6.3 Contract features.-

6.3.1 Ordering data.- Procurement documents should specify:

(a) Exact title of the specification
(b) Number of the specification
(c) Date of the specification
(d) Preproduction sample
(e) Lot size
(f) Total quantity of order
(g) Selection of applicable levels of packaging and packing

6.3.2 Inspection level.- The inspection levels and normal or tightened inspection at the start of contract shall be determined by the procuring activity.

6.4 Process improvement.- Corrective action taken to reduce rejection of component parts during processing shall be included as part of the test report when component parts rejection rate becomes excessive.

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